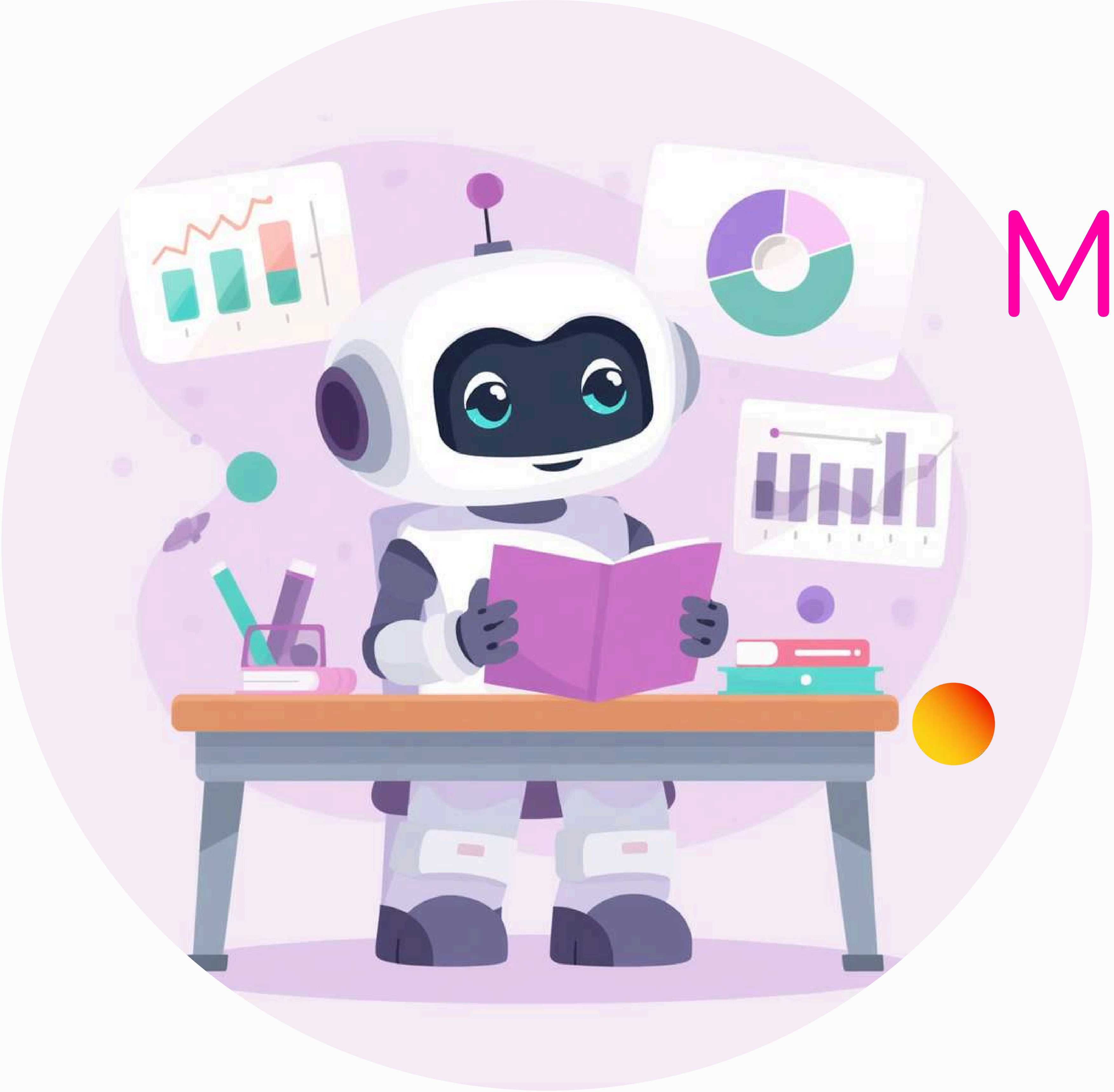
Machine Learning with scikit-learn Model

Grade 6 AI & Digital Subject
Learn how computers can learn from data and make smart predictions!

What is Artificial Intelligence?

AI means teaching computers to think and learn like humans!
Examples of AI in daily life:

- Voice assistants like Alexa and Google
- Smart games that learn how you play
- Apps that recognize your face



What is Machine Learning?

Machine Learning helps computers learn from data without being told exactly what to do.

Think of it like learning at school — you learn from many examples until you understand the pattern!

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Why is Machine Learning Important?

Video Recommendations
ML helps suggest your favorite videos and songs based on what you like to watch!

Healthcare Heroes
Doctors use ML to find diseases early and help patients get better faster!

Self-Driving Cars
ML helps cars see the road and drive safely without a human driver!

What is a Machine Learning Model?

A model is like a smart brain that learns patterns from data!
Think of it as a brain trained to recognize things — just like how you learn to tell apart different animals or colors.

How Does ML Actually Work?

Collect & Teach

Gather data and feed it to your model so it can learn patterns and rules.

Test the Learning

Check how well your model learned by testing it with new examples.

Make Predictions

Use your trained model to predict answers for things it has never seen before!

Meet scikit-learn Library

scikit-learn is a friendly tool that helps us build Machine Learning models easily!
It's a popular Python library used by coders.

With scikit-learn, you can teach computers to learn patterns from data and make smart predictions — just like magic!

Setting Up Our Coding Environment

Install Python

Download Python from python.org and install it on your computer. It's free and easy!

Install scikit-learn

Open terminal and type: pip install scikit-learn. This adds our ML tool!

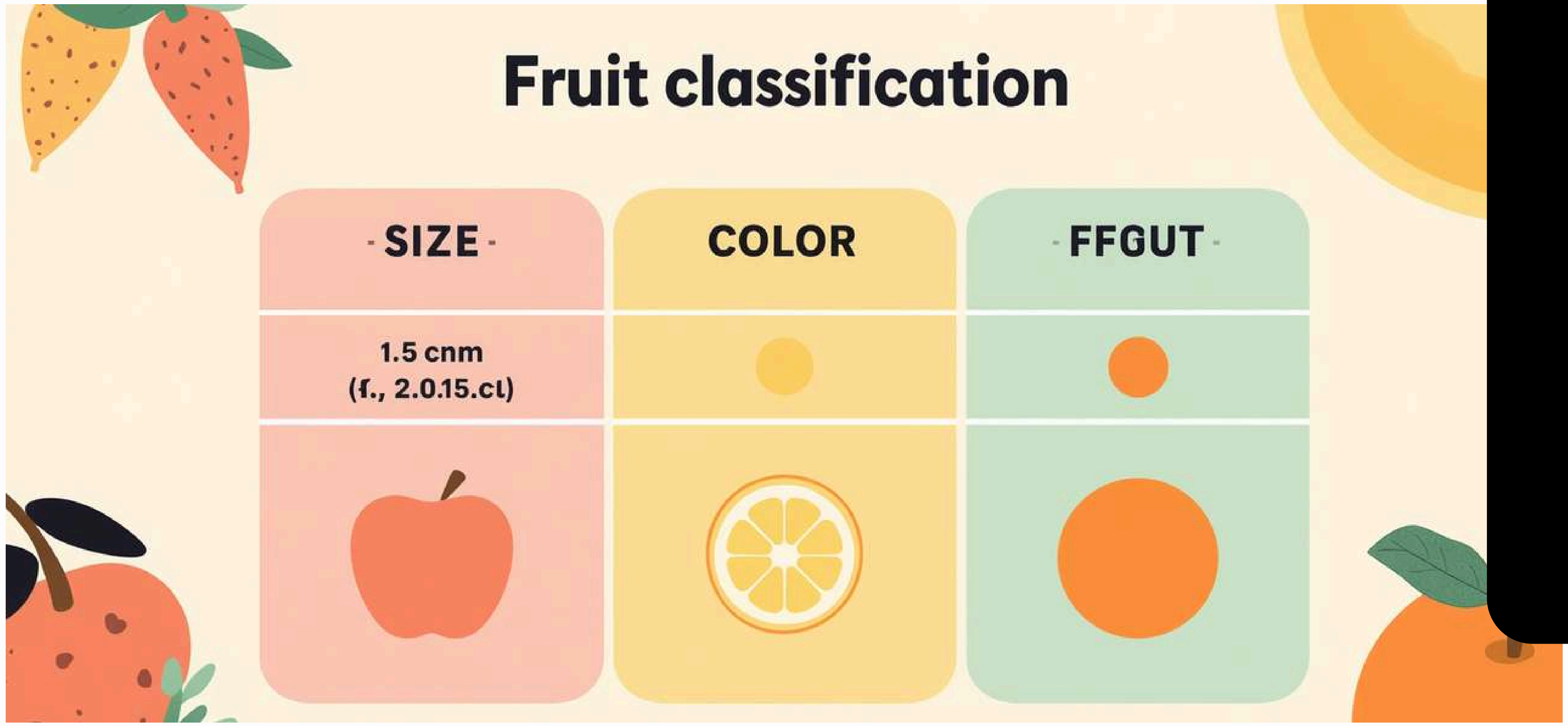
Use Jupyter Notebook

Jupyter is a friendly place to write and run Python code. Try Google Colab online!

Our Simple ML Example

We will build a model to guess if a fruit is an apple or an orange based on size and color! This fun project will help us understand how machine learning makes predictions from simple data.

Understanding Our Data



Size (cm)

How big is the fruit? This is one of our features - the input data we give to our model.

Color (scale)

Is the fruit more red or orange? This is another feature that helps our model learn.

Fruit Type

Apple or Orange? This is the label - what we want our model to predict!

Features = Input | Labels = Output

Choosing a Model Type: Decision Tree

We will use a Decision Tree — it makes decisions like a flowchart! It asks yes/no questions to sort things into groups. Example: Is the fruit big? Yes → Apple! No → Orange! Simple and easy to follow.

Writing the Code

Step-by- Step

Import & Prepare Data

Import scikit-learn library and set up your fruit data with size and color values.

Create & Train Model

Make a Decision Tree model and teach it using your fruit examples.

Predict New Fruit

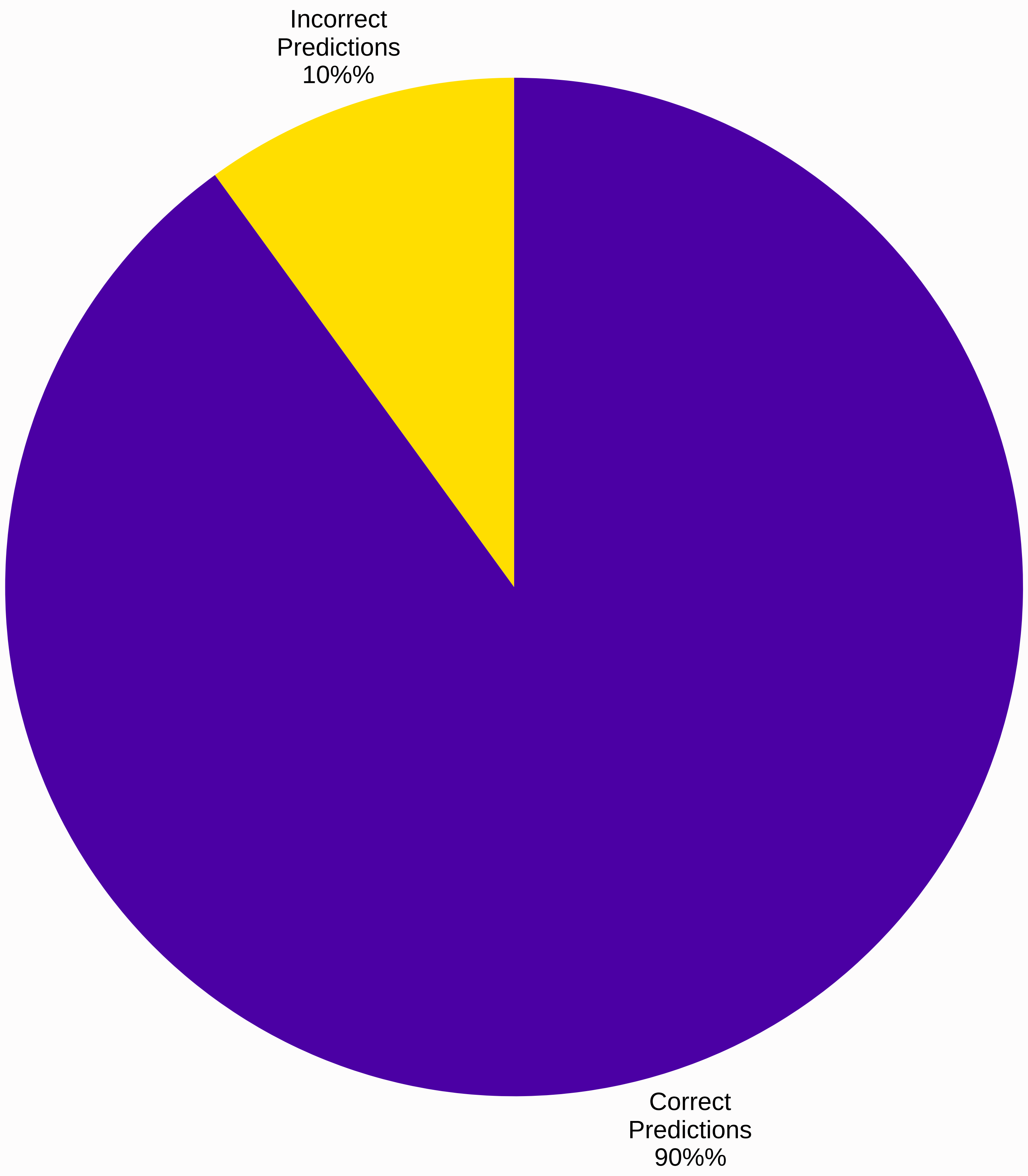
Give your model new fruit data and watch it guess apple or orange!

Running Our Model& Results

Our model guessed the fruit correctly! When we run the code, it shows the prediction result. This means our Decision Tree learned the patterns well!

Understanding the Results

Accuracy tells us how often the model is right! If our model has 90% accuracy, it means it guessed correctly 9 out of 10 times. The higher the accuracy, the smarter our model has become at predicting!



Source: AI-generated data. Replace or verify before use.

What Did We Learn Today?

ML Learns from Data

Machine Learning helps computers learn patterns from examples, just like you learn at school!

scikit-learn is Easy

This friendly Python library makes building ML models simple and fun for beginners!

We Built a Predictor

We created a smart model that can guess if a fruit is an apple or orange. Coding is powerful!

What's Next?

Keep Exploring!

Try New Data

Change the fruit data or add new fruits like bananas and grapes to see what happens!

Explore Other Models

Try different models like KNN or SVM to compare how they learn and predict.

Keep Practicing

The more you code and experiment with AI, the better you'll become at building smart programs!

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Thank You!
Any Questions?

We hope you enjoyed learning about Machine Learning and scikit-learn!
Keep exploring, keep coding, and keep asking questions!

Ask your teacher for help!
Happy Learning and Coding! 🎉